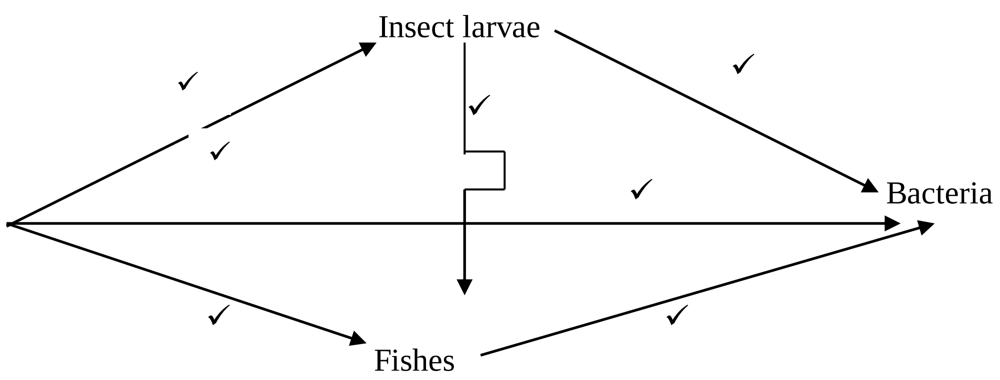
231/2
BIOLOGY
Paper 2
THEORY
Time: 2 Hours

Kenya Certificate of Secondary Education (K.C.S.E)

Marking scheme.

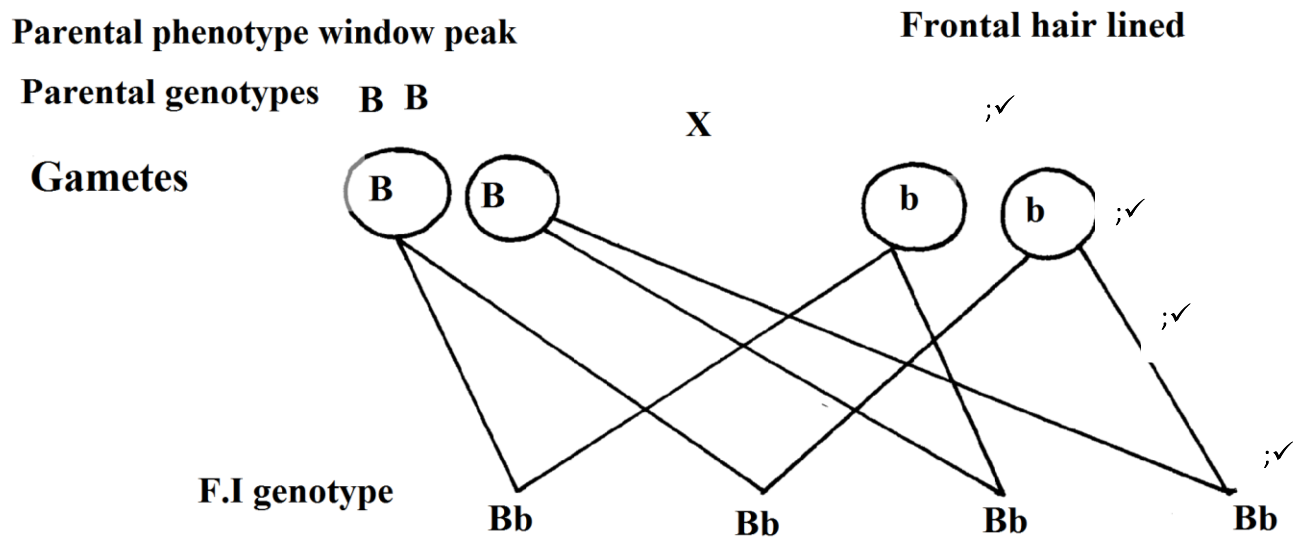
1. a) Adaptations of capillaries.
-Their walls are made up of an endothelium only which allows only part of the blood to move into the intercellular spaces;✓
-They are numerous thus creating a large S.A for exchange of materials;✓
-Have narrow lumen that maintains high blood pressure;✓
-Have sphincter muscle at the arteriole end enabling regulation of blood flow;✓
b) (i) Blood cells labeled B.
Red blood cells;✓
ii) Oxygen gas;✓
c) (i) Transport of food nutrients;✓Metabolic wastes;✓antibodies involved in defense against diseases;✓
(ii) Distribution of body heat;✓
d) (i) Pulmonary vein;✓
(ii) Hepatic vein;✓

2. (a) Where organisms in various atrophic levels don't exceed the carrying capacity;✓
(b) 500+ 1200+5000+10=7610g W 6.71kg;✓
(c) (i) Water plants;✓
(ii) Fishes;✓
(d)



(3mks)

3. (a) Gene for the window peak is dominant over gene for frontal hair lined.



Accept pun net square but genotypes must be present in order to score.

b) 1. -Gamete formation-independent assortment;✓
-Crossing over;✓
2. -Fertilization;✓
3. -Mutations;✓

c) -Tongue rolling;✓
-Sex;✓
-ABO (blood group);✓
-Free ear lobe and attached one;✓
-Pawpaw (male and female pawpaw);✓

d) Are genes located on the sex chromosomes and are transmitted together with those that are transmitted together with those that determine sex;✓

4. (a)
A-Sporangium;✓
B-Spore; ✓(rej; spores)
C-Rhizoid; ✓(rej. Rhizoids)

b) Sporulation/spore formation;✓

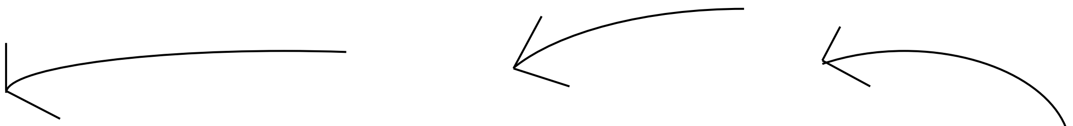
c) Absorption of water and mineral nutrients from decaying materials;✓

d) (i) Process by which female and male gamete nuclei fuse to form a diploid zygote;✓
(ii)

Ovum	Zygote
Haploid	Diploid;✓
Lower mass	Higher mass;✓

5. (a) -Sensory /Afferent Neuron;✓
-Relay/intermediate Neuron;✓
-Motor/Efferent Neuron;✓

b)



(must be drawn)

c) Grey matter;✓

d) Impulse reaching the dendrite end of relay neuron causes the synaptic vesicle to release a acetylcholine;✓(transmitter substance) that diffuse across the cliff ;✓and causes the depolarization of the motor neuron;✓

6. a) Graph C=2mks
P=2mks
A=2mks
S=1mk
<u>L=1mk</u>
8 mks.

b) i) $440cm^3 \underline{+} 5$;✓(must be shown on the graph)

ii) $\frac{440-60}{1} = 380cm^3\,perhour$;✓

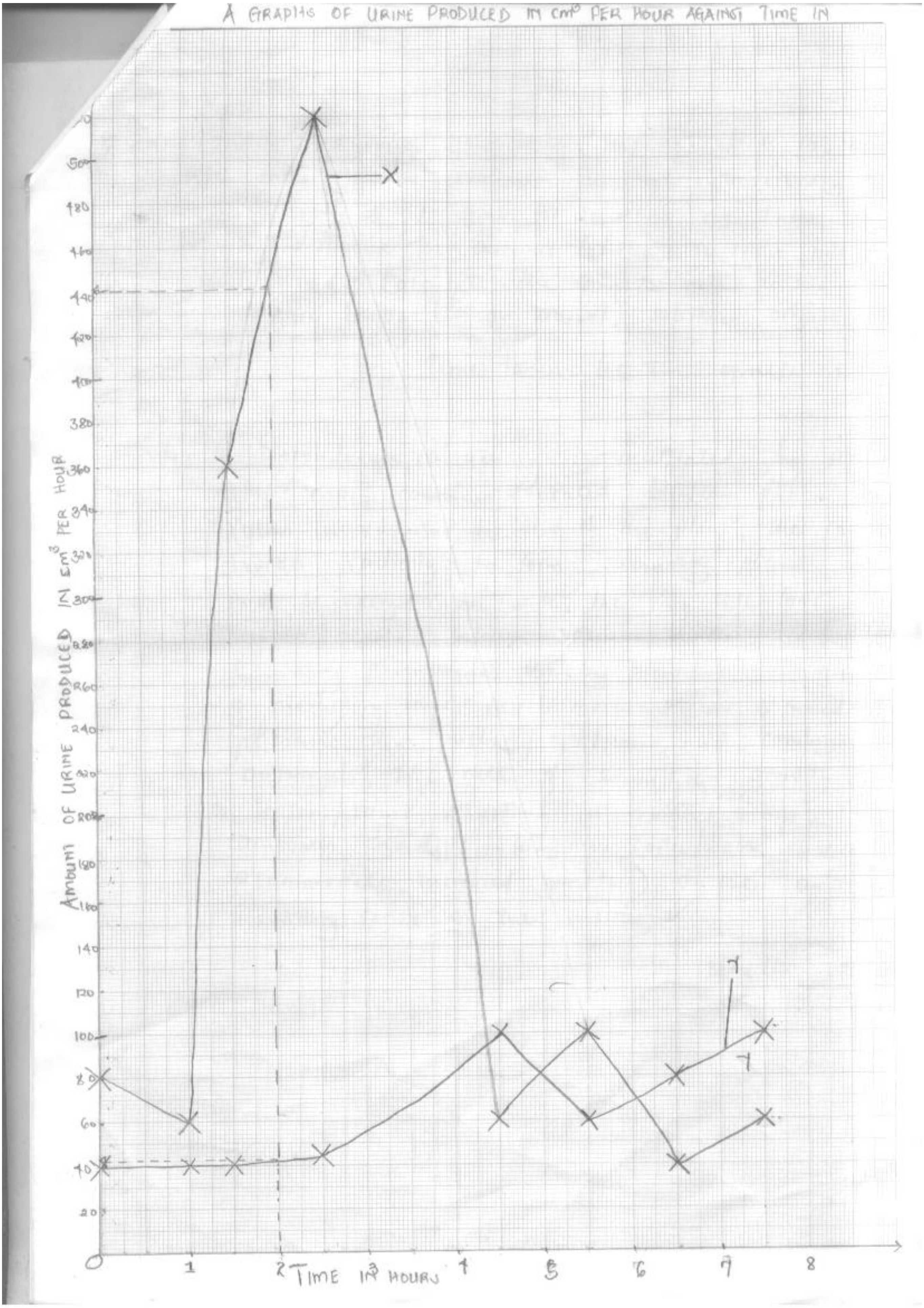
c) When osmotic pressure of blood is low due to dilution by water intake;✓rate of urine production increases;✓

d) Rate in X is higher in the first hour than in Y; because intake of water lowers conc. of blood;✓Excess is lost in urine. Therefore it reduces drastically becomes as low as in Y;✓Because excess water has been eliminated in urine and osmotic pressure of blood is normal just like X.

e) Concentration of 0.9% sodium chloride is isotonic to that of blood plasma;✓

f) The kidney is able to regulate the osmotic pressure of blood;✓

g) They are able to vary in volume of blood plasma when high due to dilution with water. The kidneys excrete excess water through urine thus lowering the volume to normal;✓


A GRAPHS OF URINE PRODUCED IN cm³ PER HOUR AGAINST TIME IN
AMOUNT OF URINE PRODUCED IN cm³ PER HOUR
TIME IN HOURS
X
Y

7. (a) Identify a young germinating seedling;✓mark its radical with Indian ink or permanent ink at intervals e.g. of 2mm;✓Leave it to grow for some time e.g. 24 hrs or overnight;✓Measure the distances between successive ink marks;✓and record;✓calculate the growth rate as;

$$\frac{\text{New length(of an interval) subtract original length}}{\text{Time taken for the growth}}$$ (max 6mks)

b) Secondary thickening is facilitated by meristematic cells;✓called cambium;✓located between phloem and xylem in vascular bundles of the plant;✓the cambium divides radially;✓to form secondary phloem outside and secondary xylem inside;✓the ring cells forms intravascular cambium/cambium between vascular bundles divides to form secondary parenchyma;✓thereby increasing the growth of modularly rays; much more xylem is formed than phloem;✓thus pushing phloem and cambium ring outwards;✓the rate of secondary growth is dependent on seasons/rains;✓resulting in annual/rings;✓cork cambium divides to form new cork/bark tissue ;✓to accommodate increased growth)on the outside and secondary corlex on the inside;✓ (max 14mks)

8. (a) Light stage

It occurs in the grana/granum of the chloroplast;✓the chlorophyll in the chloroplast traps;✓light energy;✓used to split water molecules into oxygen and hydrogen atoms;✓in the process called photolysis.

$$\left[acc2H_2O_2 \frac{Lightenergy}{chlorophyl} 4H + O_{2(g)}\right]$$ (For 3mks)

The hydrogen atoms produced enters the dark stage, while oxygen is released to the atmosphere/re-utilized by the plant in respiration;✓some of solar energy absorbed by chlorophyll molecule is used in formation of ATP,used later in the dark stage;✓

(11mks max 10mks))

b) Dark stage.

It involves combination of carbon (iv) oxide and hydrogen atoms;✓in a series of enzymes catalyzed;✓reactions to form simple sugar;✓(e.g. sucrose)in the process called carbon (iv) oxide fixation;✓the process required energy which is provided by ATP;✓(formed during the light stage);✓some of the sugars formed are directly utilized;✓by the plant cells while the rest are converted to starch;✓for storage.

Amino acids;✓and fatty acids ;✓are also formed during stage of photosynthesis;✓ (10mks)